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First Step Toward Autonomous Drilling via Drilling Process Automation and AI-Powered ROP Optimization in ADNOC Offshore Field

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Abstract

This paper details the first offshore deployment of an artificial intelligence (AI)-powered drilling automation system on a jackup rig in the UAE by ADNOC Offshore. The project marks a significant step in the digital transformation of offshore drilling, integrating both automated surface equipment control and AI-driven rate of penetration (ROP) optimization into the full drill-a-stand process (Noshi and Schuber, 2018; Bello et at., 2016). Two development wells were drilled using the system, which not only improved operational efficiency but also established a new performance benchmark for future operations. The study involved benchmarking against offset wells, applying performance diagnostics, and documenting technical and operational lessons learned. By demonstrating measurable performance improvements in ROP and connection time reductions, this paper supports the viability of full-cycle drilling automation in offshore environments. The trial validates the role of intelligent automation in ADNOC's digital roadmap and offers a replicable blueprint for broader fleet-wide implementation. The results also underscore the importance of cross-functional collaboration and proactive crew engagement in realizing technology benefits. This work contributes new data and practical insights to the body of knowledge surrounding automation and AI in drilling operations, particularly in complex offshore settings where real-time decision-making and execution consistency are critical to success.

Introduction

The energy sector continues to seek avenues for cost reduction and performance enhancement amid volatile market dynamics and increasing environmental regulations. As exploration and production shift toward more technically challenging reservoirs, particularly offshore, the margin for inefficiency narrows considerably. Traditional drilling operations are often constrained by human decision-making, inconsistent execution, and delayed responses to real-time conditions (Teale, 1965). In response to these challenges, many operators are investing in digital drilling technologies, including artificial intelligence (AI) and machine learning (ML) for optimization, and robotics and control systems for process automation (Noshi and Schubert, 2018).

ADNOC Offshore has taken a proactive stance in leveraging such technologies to modernize its operations. The company recognizes that automation not only increases efficiency but also reduces exposure to health and safety risks by minimizing manual intervention (Dupriest and Koederitz, 2005). The trial discussed in this paper represents the first fully automated drill-a-stand operation performed offshore in the UAE. It integrates AI-driven rate of penetration (ROP) with automated sequencing of rig equipment, forming a closed-loop system that can execute and adapt in real-time (Bello et al., 2016). By applying these technologies, ADNOC aims to enhance drilling consistency, minimize invisible lost time (ILT), and foster a data-centric culture onboard its rigs. This paper describes the trial's design, execution, and outcomes in depth.

Objectives and Scope

The primary objective of the automation trial was to evaluate the technical and operational feasibility of integrating AI and surface automation into the offshore drilling process, specifically focusing on automating the drill-a-stand cycle and the on-bottom ROP. This included testing the ability of the AI system to optimize drilling parameters dynamically and the capacity of the surface machinery to execute sequences with precision and repeatability. The scope of the pilot was intentionally focused to maximize control and comparability. It included:

- Deploying the solution on a single jackup rig to reduce operational variability.
- Selecting two wells with consistent lithological characteristics to ensure data relevance.
- Benchmarking against five offset wells drilled with conventional manual processes in the same field.
- Measuring and analyzing both on-bottom ROP performance and the time associated with pre- and post-connection phases.

Moreover, the project included a structured review of system usability, crew feedback, system utilization rates, and operational risks. The collected data and operator experience will be used to define implementation strategies for additional rigs, including upskilling requirements, performance KPIs, and risk mitigation plans.

Methodology

The implementation of the automation trial followed a structured, five-phase methodology designed to ensure both operational readiness and analytical rigor. These phases included: rig selection and preparation, system integration, performance benchmarking, live deployment, and post-trial evaluation (Noshi and Schubert, 2018).

Rig Selection and Preparation

The first phase involved identifying a rig that met all technical and logistical prerequisites for automation. Selection criteria included compatibility with digital infrastructure, alignment with ADNOC's operational schedule, and the availability of leadership support onboard. The selected jackup rig already featured modern control systems and networked instrumentation, enabling seamless integration of new automation components. Prior to deployment, a series of pre-trial audits were conducted to validate sensor accuracy, verify network stability, and ensure that all equipment met calibration standards.

System Integration

Once the rig was selected, the automation systems were installed and interfaced with existing rig infrastructure. The system architecture comprised two core elements: a surface control module responsible for automating mechanical sequences during connections, and an AI-powered ROP optimizer that adjusted drilling parameters in real time. The optimization engine utilized reinforcement learning to predict optimal weight on bit (WOB) and rotary speed (RPM) combinations. Control commands were transmitted directly to rig machinery, with all parameter changes confined to pre-approved safety and performance envelopes.

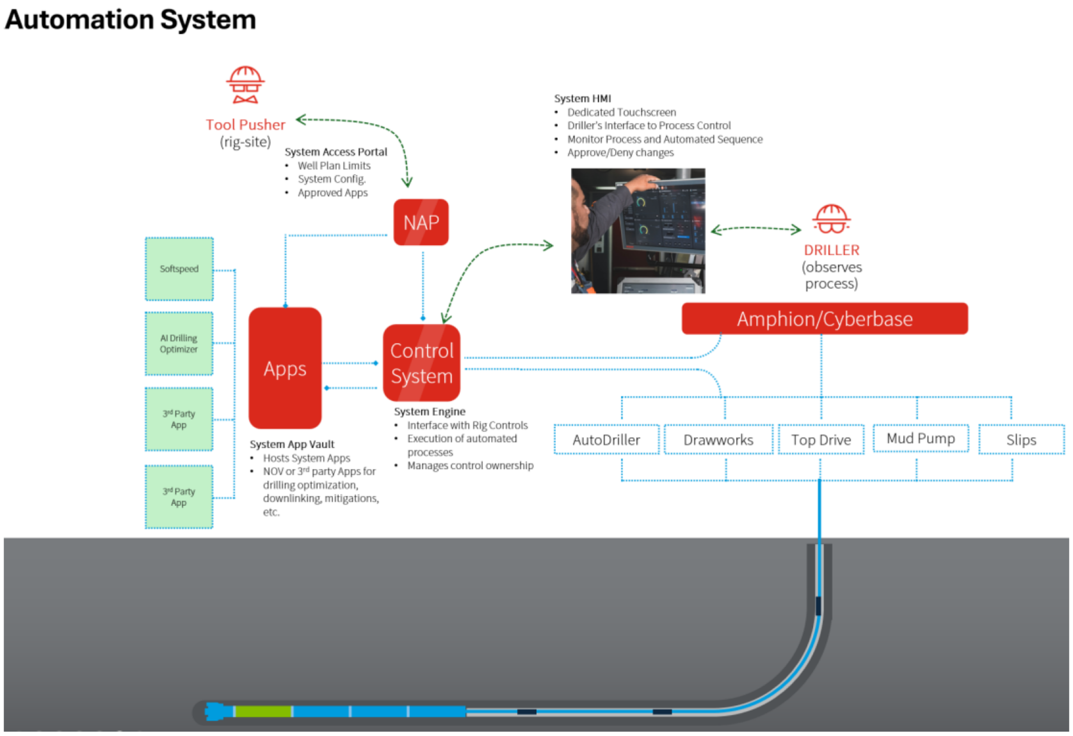

Figure 1—Automation System

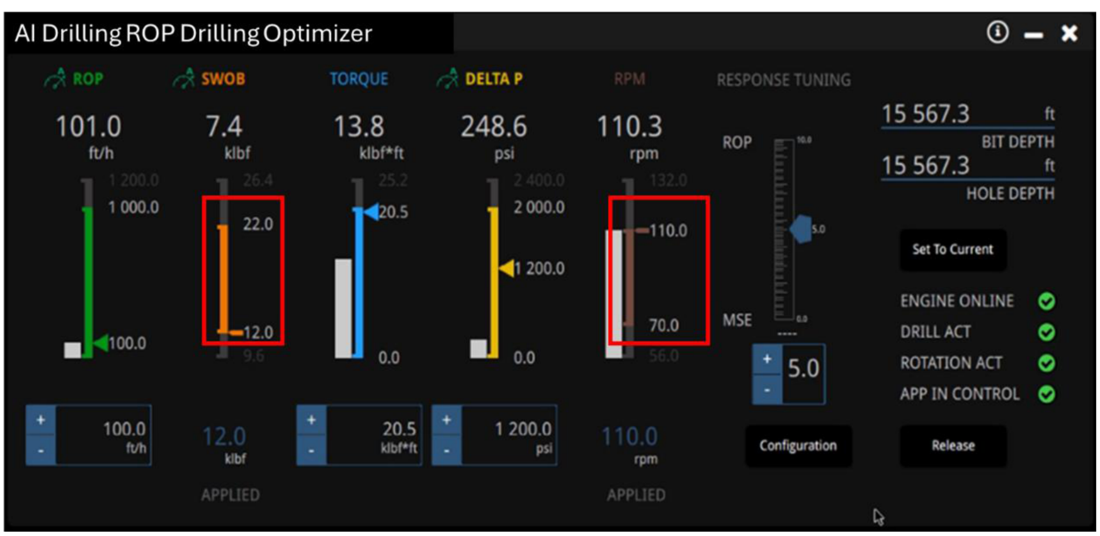

Figure 2—Parameter windows Input for WOB and RPM

AI-Powered ROP Optimizer

The AI-powered Rate of Penetration (ROP) optimizer integrates six distinct algorithms, with the core component being a dynamic learning algorithm. These algorithms span from advanced machine learning models to deterministic analytical trend analysis, each contributing uniquely to the optimization process.

The system architecture and functional roles of each algorithm are outlined below, highlighting their respective contributions to overall performance enhancement.

$$\boxed{OverallPerformance = f(ROP, RPM, SWOB, deltaP, dysfunction, MWD\ Shock\ and\ Vibrations, ECD, inclination, ..)}$$

$$\boxed{Friction\ Factor = f(hookload, SWOB, TopDrive\ Torque, RPM, ..)}$$

$$\boxed{ECD = f(Physics\ Model - Inferred, PWD, WOB, ROP, RPM)}$$

Figure 3—Algorithm functions

Data Flow and Control Strategy

The optimizer ingests real-time operational input data, as illustrated in the input dataset below. Among all parameters, Weight on Bit (WOB) and Rotary Speed (RPM) are the primary control variables, actively adjusted via the control system through the optimization interface.

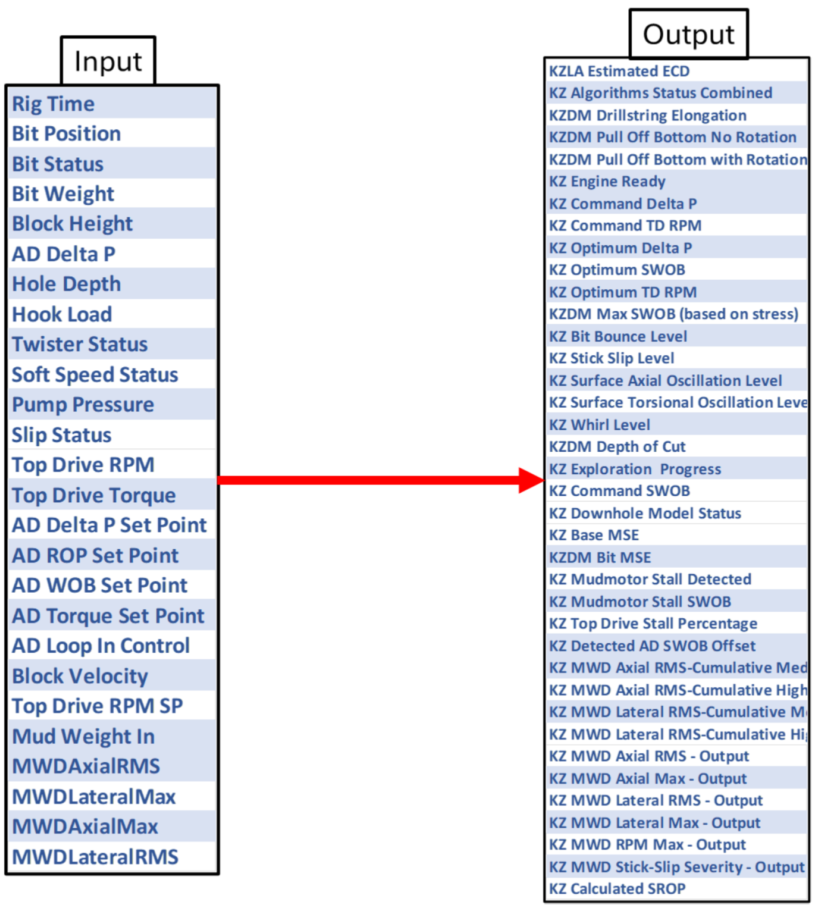

Figure 4—Input and Output variables

Dysfunction Handling and Performance Gains

When deployed in conjunction with stick-slip mitigation technology, the AI optimizer is capable of disregarding stick-slip effects in its optimization model while simultaneously mitigating other drilling dysfunctions (e.g., bit bounce, whirl, and vibration). This decoupling allows the optimizer to focus on maximizing ROP without being constrained by stick-slip disturbances, resulting in a more robust and performance-driven drilling operation.

Performance Benchmarking

To objectively assess the impact of automation, a performance baseline was established using five historical wells drilled in the same field. These offset wells provided data on key performance indicators, including average ROP, pre- and post-connection times, and total drilling duration per section. Pre-connection is the time from the time you stop drilling and go into slips, and post connections is the time you go from slips to tagging bottom. This baseline enabled a side-by-side comparison with trial results, allowing the team to isolate the contribution of automation to any observed performance gains.

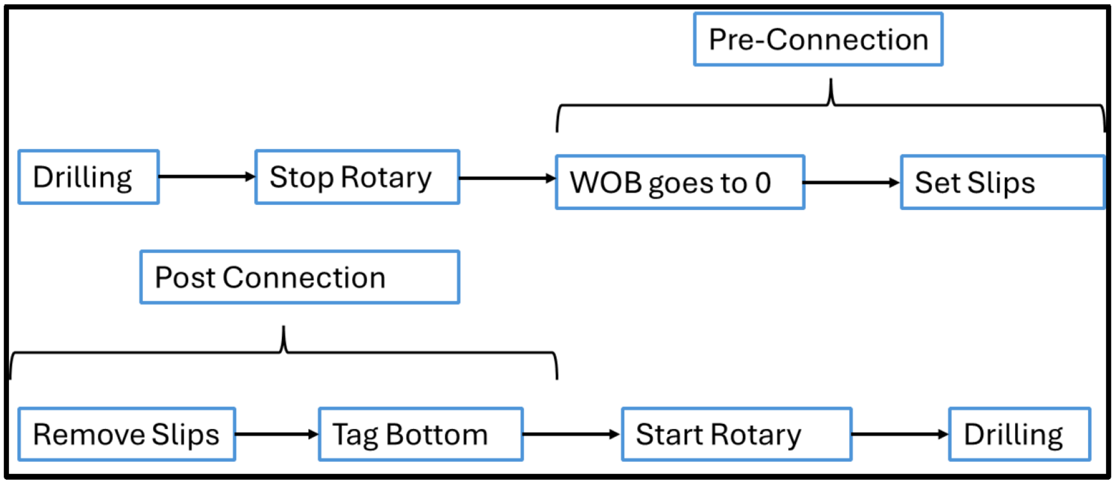

Figure 5—Pre-Connection and Post Connection Example

Field Trial Deployment

The automation system was deployed during the drilling of two development wells. These wells included a 12¼-in. intermediate section and an 8½-in. lateral section, both selected for their geological similarity to prior wells in the benchmarking set. The system operated continuously during all drill-a-stand activities, including on-bottom optimization and automated connection handling. Field engineers monitored system performance in real time using live dashboards and event-logging tools, ensuring traceability and immediate feedback for any anomalies encountered during the operation.

Post-Trial Evaluation

After drilling activities were completed, the trial entered the evaluation phase. Using analytics software, the team reviewed detailed time breakdowns, ROP trends, and connection performance across both trial and benchmark wells. Connection sequences were analyzed down to individual machine actions to identify micro-efficiencies. In addition, system utilization rates and manual override frequencies were recorded to assess user adoption and reliability. Feedback from rig crews and drilling engineers was collected to inform potential refinements for future deployments.

Results and Observations

The automation trial demonstrated clear operational improvements across multiple performance dimensions, validating the impact of intelligent systems on offshore drilling efficiency. As Shown in Table 1, both quantitative performance metrics and qualitative operational insights underscore the value of AI-enabled automation for real-time decision-making and consistent execution.

Table 1—Performance Highlights

Metric	Baseline	Trial Result	Improvement
12¼" in. Section Average ROP	18.5 ft/hr	23.5 ft/hr	+27% ROP improvement
8½" in. Section Average ROP	22.0 ft/hr	25.3 ft/hr	+15% ROP improvement
12¼" in. Pre-Connection Time	9.3 minutes	5.9 minutes	−37% time reduction
8½" in. Pre-Connection Time	7.5 minutes	6.0 minutes	−20% time reduction
12¼" in. Post-Connection Time	(not provided)	(calculated internally)	−35% time reduction
8½" in. Post-Connection Time	(not provided)	(calculated internally)	−21% time reduction
System Utilization Rate	—	94%	High trust and consistency
Manual Overrides (by exception)	—	<3%	Isolated to anomaly handling
NPT Linked to Automation	—	0 hours	Zero attributable downtime

Improved Drilling Efficiency (ROP Gains)

The AI-driven ROP optimization engine led to substantial improvements in drilling speed. In the 12¼" in. intermediate section, the system increased the average ROP by 27%—from 18.5 ft/hr in offset wells to 23.5 ft/hr. Similarly, the 8½" in. production section showed a 15% ROP increase, rising from 22 ft/hr to 25.3 ft/hr. These gains were not isolated but observed consistently across multiple stands, reinforcing the optimizer's ability to identify high-performance operating windows for WOB and RPM without the need for manual drill-off tests.

Optimization was achieved by enabling the AI-driven ROP optimizer to operate within dynamic parameter windows rather than fixed set points, as used in conventional autodrillers. This concept is consistent with early work on mechanical specific energy (MSE) and its role in drilling efficiency (Teale, 1965; Dupriest and Koederitz, 2005). Typical ranges included 10–15 klbf between minimum and maximum WOB and 20–30 RPM between lower and upper speed limits. This methodology allowed continuous identification of high-performance operating conditions without manual drill-off tests. A key limitation, however, is the system's lack of lithology awareness; in abrasive formations such as sandstones, lower parameters are often required to mitigate bit wear, making ROP maximization less desirable.

Connection Time Reductions

Automation of the drill-a-stand process, particularly the pre- and post-connection sequences, led to significant time savings. In the 12¼" in. section, pre-connection time was reduced by 37% (from 9.3 to 5.9 minutes), while post-connection sequences improved by 35%. In the 8½" in. section, the pre-connection sequence improved by 20%, and post-connection by 21%. These reductions were primarily attributed to the system's precise sequencing of equipment operations and reduced variability in execution.

High System Utilization and Operator Trust

The system was utilized in 94% of eligible drilling cycles, reflecting high confidence among rig personnel. Eligible drilling cycles is defined by only rotary drilling and is not utilized during sliding, when the top drive is not being utilized. Only a small percentage of stands—fewer than 3%—required manual intervention, generally due to isolated anomalies such as downhole vibration spikes or sensor inconsistencies.

Operational Reliability and Stability

Throughout the trial, no non-productive time (NPT) was linked to the automation system. Drilling remained within safe operational windows for torque, equivalent circulating density (ECD), and downhole vibration.

These outcomes suggest the system is not only effective but also dependable across varied drilling conditions, validating its suitability for broader deployment.

Stick-Slip Mitigation

A key feature of the trial was the deployment of an advanced stick-slip mitigation system during the rotary drilling phase. Stick-slip, a common form of torsional vibration, occurs when the bit alternates between sticking and suddenly rotating, leading to excessive torque fluctuations and mechanical wear. The mitigation system utilized a dynamic surface control strategy that modulated top drive RPM in response to measured torque oscillations. By introducing oscillatory variations in surface RPM—synchronized with the torque wave—the system effectively dampened the stick-slip cycle. This proactive modulation created a destructive interference pattern that neutralized the buildup of torque, stabilizing the bit's rotational speed. The technology proved particularly effective in formations prone to severe torsional instability. When active, the system significantly reduced vibration intensity, minimized cutter damage risk, and extended tool life. Real-time diagnostics also enabled operators to monitor severity levels and system response, providing a foundation for further refinements. The inclusion of this system contributed directly to improved ROP stability, longer bit runs, and a smoother overall drilling operation. It exemplified how intelligent surface interventions can address downhole dysfunctions without necessitating additional hardware or complex interventions.

Digital Twin

The concept of the digital twin refers to a real-time, physics-based model of the drilling system that mirrors operational behavior and provides predictive insights. In broader industries, digital twins have been used to enable full lifecycle management of assets through continuous synchronization of physical and digital states (Grieves, 2006). This virtual representation runs in parallel with live drilling operations, leveraging real-time data to simulate downhole conditions and forecast potential dysfunctions. In the context of this trial, the digital twin was integrated with the Intelligent Drilling Optimizer (IDO) /AI-powered ROP Optimizer and used to augment decision-making during rotary drilling.

The model encompassed several key components: The model encompasses a full Finite Element Analysis (FEA) to model torque and drag, stress and buckling, and axial loads in the drill string, vibration modeling to avoid resonant frequencies that could trigger axial, torsional, or lateral vibration, and synthetic modeling of WOB and RPM behavior to optimize bit performance. The system also incorporated mud motor stall detection logic, identifying damaging parameter combinations and adapting accordingly.

The benefit of this setup was twofold: it reduced the need for time-consuming post-well analyses and allowed the optimization system to avoid known dysfunctions in real time. Ultimately, the digital twin served as a continuous learning environment for the AI, improving the optimizer's adaptability and safety under varying drilling conditions.

While the trial successfully met its performance objectives in terms of efficiency gains and system reliability, several important challenges emerged during implementation. These challenges were both technical—related to the dynamic nature of the downhole environment—and human-lefted, highlighting the importance of operational alignment and cultural readiness for digital transformation.

Hole Cleaning and ECD Management

One of the early operational challenges involved maintaining effective hole cleaning as drilling speeds increased. Although the optimization algorithm excelled at maximizing ROP, it initially operated without integrated constraints related to cuttings transport and ECD. As a result, elevated ECD levels were observed —particularly in deviated and horizontal intervals—posing a potential risk to wellbore stability and pressure management.

To address this, the system was enhanced with a real-time ECD management layer that combined both calculated and measured data. Specifically, algorithms were deployed to model the downhole ECD profile based on drilling parameters, generating a synthetic ECD output in real time. This modeled ECD was then overlayed onto the actual sensor data transmitted from the Pressure While Drilling (PWD) tool, providing a dual-layer validation mechanism. With this combined dataset, the AI engine could intelligently adjust ROP in response to dynamic ECD trends—proactively maintaining safe operating windows without requiring manual intervention.

The ECD-estimation algorithm couples a physics-based hydraulics model with a real-time learning layer. With each new ECD datapoint, it recalibrates its parameters against surface drilling variables, hole depth, mud-flow rate, and other inputs, then infers the equivalent circulating density at the bit (synthetic ECD). Because the model includes the added pressure drop from the ECD sensor down to the bit, synthetic ECD values typically read higher than the raw MWD measurements.

Learning and inference occur continuously, and the synthetic ECD output is refreshed twice per minute. This synthetic ECD value is then used along with other KZ learnings to estimate drilling parameters that keep ECD safely below the given threshold.

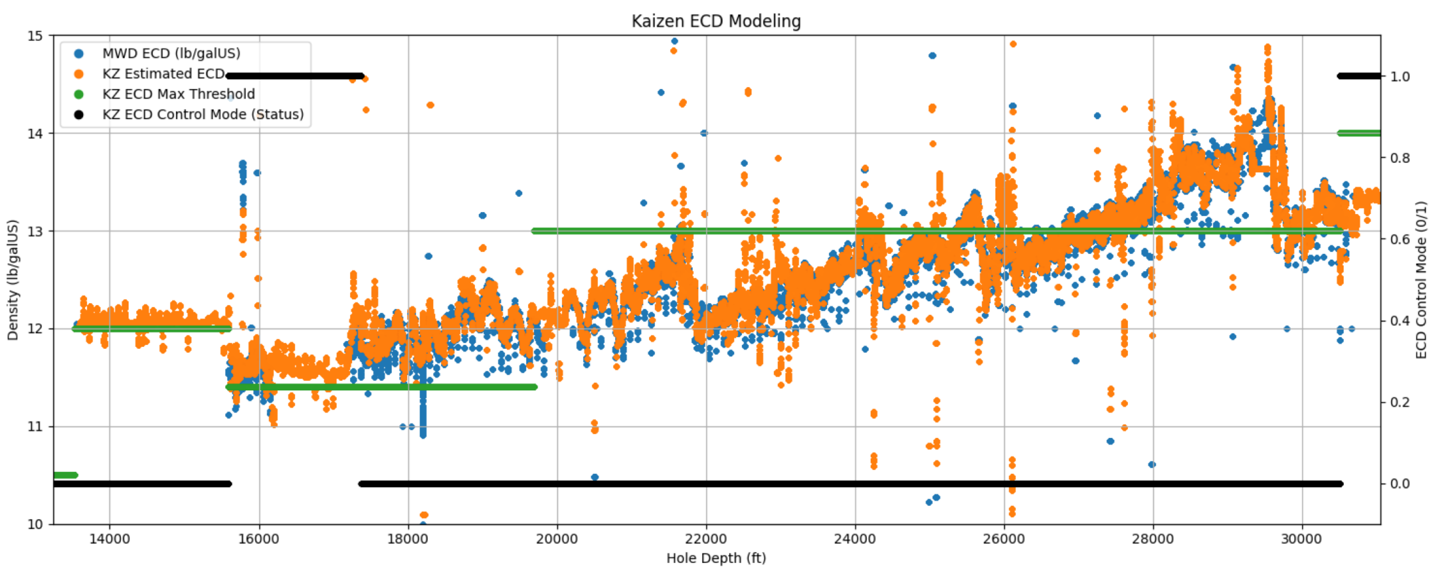

Chart 1—Measure ECD vs Modeled ECD

The corresponding graph illustrates a consistent delta between the modeled ECD and the measured ECD from the PWD tool, with the modeled values trending approximately 0.1 to 0.2 ppg higher. This difference is intentional and expected, as the model is designed to predict ECD conditions specifically at the bit, whereas the PWD measurements reflect conditions slightly uphole. By forecasting ECD at the bit location—where pressure conditions are most critical—the model enables the optimizer to make more conservative and proactive adjustments, thereby safeguarding against potential downhole instability before it manifests in the measured data. This predictive capability enhances the robustness of the optimization loop and supports better decision-making in real-time operations.

This advancement reinforced an important operational insight: maximizing ROP must always be balanced against the preservation of hole quality. Integrating real-time ECD modeling and monitoring into the optimization framework ensures that performance gains are achieved without compromising wellbore integrity.

Formation Transitions and Geomechanical Complexity

Geological complexity introduced a second layer of challenge, particularly in intervals with interbedded lithologies. These transitions often induced abrupt changes in drilling dynamics, evidenced by torque spikes, and vibration signatures.

To address this, operators defined "no-optimization zones" ahead of time based on formation logs and offset well experiences. These zones temporarily disabled the optimizer and reverted control to conservativea, pre-set parameters. In tandem, a formation-specific setpoint roadmap was created to guide the system when operating near known trouble zones.

Human Factors and Crew Acceptance

A pivotal aspect of the trial's success was the engagement of the rig crew and their integration into the digital transformation journey. While the automation system delivered advanced capabilities in decision support and performance optimization, its effectiveness was greatly enhanced by strong collaboration with the human operators onboard.

ADNOC Offshore recognized the importance of fostering a confident and informed workforce and therefore introduced a comprehensive change management strategy. This included pre-deployment training sessions, collaborative operational planning with rig leadership, and open communication of system goals and real-time performance metrics. Throughout the trial, intuitive dashboards and visual feedback tools provided operators with clear insights into the optimizer's decision-making process and tangible benefits in drilling efficiency.

As the operation progressed, the crew's familiarity with the system grew, leading to increased trust in its reliability and consistency. Rather than replacing the judgment of experienced personnel, the technology was embraced as a complementary tool that enhanced decision-making and supported operational excellence.

Continuous engagement and feedback from the field team played a key role in refining system behavior. This participatory approach created a feedback loop that strengthened both system performance and operator confidence, paving the way for broader adoption and alignment between human expertise and intelligent automation.

System Calibration and Iterative Tuning

Deploying an advanced AI system in a live offshore environment revealed several nuances in system behavior. Early trial phases uncovered edge-case conditions where the optimizer's aggressiveness led to suboptimal performance or risk exposure.

Several firmware updates and field-level software patches resolved the issues. Importantly, ADNOC maintained a field engineering team onsite during critical phases of the deployment, allowing rapid response and iterative fine-tuning. Lessons from these updates were incorporated into a continuous improvement cycle, improving the robustness of future deployments.

From a systems engineering perspective, the trial highlighted the importance of flexible model tuning parameters, robust anomaly detection logic, and rollback capabilities in autonomous drilling systems. Monitoring tools capable of alerting operators to confidence levels and decision boundaries also proved essential in balancing trust with oversight.

The user defines the target parameter, whether prioritizing rate of penetration (ROP) or overall efficiency, and the system internally adjusts its control variables to align with the expected operating conditions. An embedded self-tuning algorithm continuously optimizes configuration to ensure proper system response, minimizing the need for exact manual calibration. As a result, the operator specifies only the desired operational focus while the system autonomously converges on the appropriate tuning range, as illustrated in Figure 6.

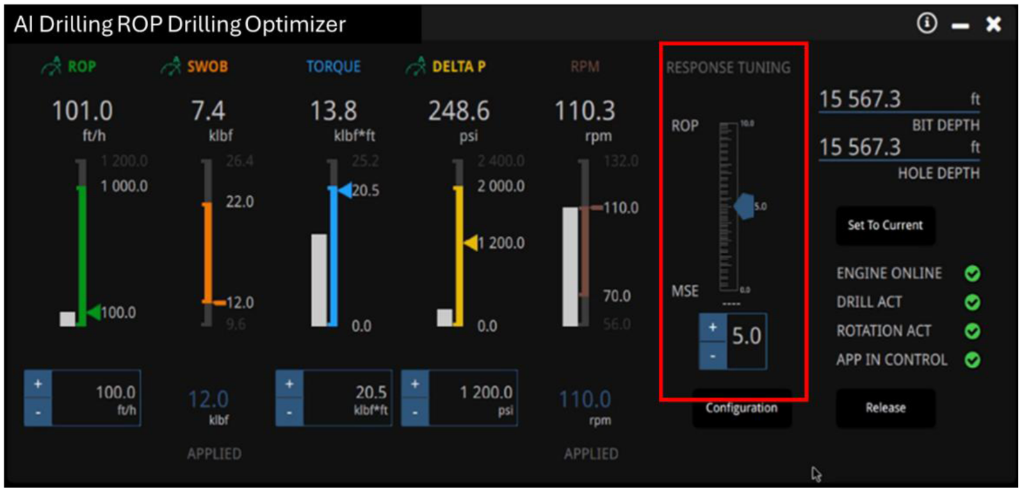

Figure 6—Response Tuning

Conclusions

This trial validates that AI-powered drilling automation systems can significantly improve offshore drilling performance (Bello et al., 2016). It delivers measurable gains in ROP and connection efficiency while providing operational consistency and safety improvements. The results also highlight that successful implementation is as much about people as it is about technology. Crew engagement, training, and support are critical to maximizing system potential.

The trial also serves as a launchpad for scaling this technology across ADNOC's fleet. Broader integration is envisioned as part of ADNOC's long-term digital transformation journey, enabled by remote support capabilities, advanced crew development programs, and alignment with predictive maintenance strategies.

Importantly, this work makes several novel contributions to the petroleum engineering literature. It demonstrates the real-time integration of ROP optimization with mechanical automation, quantifies the impact on connection cycle consistency and drilling efficiency across varied lithologies, and introduces a framework for optimizing human-system interaction. In addition, the trial establishes a scalable deployment model for Jack-up fleets, offering a replicable pathway for other offshore operators.

In sum, ADNOC's first offshore automation trial not only confirms the business case for digital transformation in drilling but also provides unique insights and frameworks that can guide global efforts in digital drilling, intelligent automation, and human-technology integration. Together, the combination of real-time AI optimization, surface automation, and structured deployment methodology forms a powerful platform for operational excellence in complex environments.

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